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Bamboo Sticks as Reinforcement in Concrete: A Research Review from Interface Bonding to Structural Performance

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Abstract

As a representative of sustainable building materials, bamboo has regained research attention in the field of concrete reinforcement. This paper systematically reviews the research progress of bamboo reinforced concrete (BRC), providing a comprehensive analysis from material properties and interface bonding mechanisms to structural member performance. Studies show that bamboo tensile strength can reach 3800 kg/cm², but its elastic modulus is only one-tenth that of steel, resulting in bamboo reinforced members exhibiting large deflection and wide cracks. Weak interface bonding is the core bottleneck restricting bamboo reinforcement applications. Epoxy coating treatment can increase bond strength by 5.29 times, and rubber coating can significantly improve initial slip behavior. The flexural load-bearing capacity of bamboo reinforced concrete members is approximately 46% that of steel reinforced concrete, but hybrid reinforcement design (steel-bamboo combination) can achieve mechanical efficiency close to that of purely steel-reinforced beams. Regarding fiber incorporation, 3% bamboo fiber content can increase compressive strength to over 300 kg/cm², but at high content levels, compressive strength shows a declining trend. Current research consensus holds that bamboo reinforcement is more suitable for low-rise buildings, temporary structures, and non-critical load-bearing members, and its application requires adequate protective treatment and rational design methods as prerequisites.

Keywords: bamboo reinforced concrete; interface bonding; sustainable construction; hybrid reinforcement; mechanical properties

1. Introduction

As the most consumed building material globally, concrete production accounts for approximately 8% of total global carbon emissions. Against this backdrop, seeking renewable, low-carbon alternative building materials has become an important issue in civil engineering. Bamboo has attracted attention due to its short growth cycle (3-8 years to reach maximum strength), high specific strength, and abundant resources. Research on bamboo as concrete reinforcement material can be traced back to the 1970s, when the classic study by Geymayer and Cox published in 1970 systematically explored the feasibility of bamboo reinforcement as temporary structural reinforcement, pointing out that the

ultimate flexural capacity of bamboo reinforced concrete members can reach 2 to 4 times that of plain concrete.

However, research on bamboo reinforced concrete has not been smooth sailing. The interface bonding problem between bamboo and concrete, durability concerns regarding bamboo, and the variability of bamboo mechanical properties have long constrained the engineering application of this technology. In recent years, with the deepening of sustainable building concepts and advances in material treatment technology, bamboo reinforced concrete research has shown a revival trend. Between 2023 and 2025, several high-level research papers explored the influence mechanisms of different forms of bamboo reinforcement (bamboo strips, bamboo bars, bamboo fibers) on concrete performance.

This paper aims to systematically review the current research status of bamboo reinforced concrete, focusing on the following questions: What are the material characteristics of bamboo as a concrete reinforcement? What are the solution paths for interface bonding problems? What level of structural performance have bamboo reinforced concrete members achieved? What are the core challenges and future directions facing current research?

2. Material Characteristics and Reinforcement Feasibility of Bamboo

2.1 Mechanical Properties

The mechanical properties of bamboo exhibit significant anisotropic characteristics, with tensile strength in the grain direction far exceeding that in the transverse direction. Early research by Geymayer and Cox reported that bamboo tensile strength can reach as high as 54,000 psi (approximately 3800 kg/cm²), a value comparable to medium-low carbon steel. However, bamboo's elastic modulus is only about one-tenth that of steel. This fundamental difference determines that bamboo reinforced concrete members exhibit deformation characteristics entirely different from steel reinforced concrete when subjected to forces: members tend to produce large deflections and wide cracks.

Recent research has further refined the data on bamboo mechanical properties. Sayeed et al. (2023) conducted tensile tests on bamboo bars made from Moso bamboo (Phyllostachys pubescens), finding that all specimens failed due to bamboo fiber tearing, with failure locations concentrated in the middle or at end anchorage zones. The diameter of bamboo bars (1 mm, 1.5 mm, 2 mm) and length-to-diameter ratio (10, 20, 30) were proven to be key parameters affecting their reinforcement effectiveness.

2.2 Compatibility with Concrete

The core issue of bamboo as concrete reinforcement lies in the interface compatibility between the two materials. Bamboo has the characteristics of water absorption and swelling, and shrinkage upon drying. During the hydration process after concrete pouring, bamboo reinforcement absorbs water and expands,

then shrinks during curing and drying. This process destroys the originally formed bonding interface. Additionally, the alkaline environment of concrete pore solution (pH approximately 12.5-13.5) has a degrading effect on lignin and hemicellulose in bamboo, and long-term exposure may lead to degradation of bamboo mechanical properties.

Recent research conducted by Japanese scholars specifically examined the alkali resistance of bamboo reinforcement embedded in concrete, confirming the performance changes of bamboo in alkaline environments through experiments, and based on this, conducted experimental and analytical research on the ultimate flexural strength of bamboo reinforced concrete slabs. This work provides basic data for durability design of bamboo reinforced concrete.

3. Improvement Strategies for Interface Bonding Performance

Weak interface bonding is the core challenge in bamboo reinforced concrete research. The bond strength between untreated bamboo reinforcement and concrete is far lower than that of steel reinforcement, which is the key reason causing premature slip and cracking in bamboo reinforced concrete members under load. To address this problem, researchers have developed three main categories of improvement strategies.

3.1 Surface Treatment and Coating

Surface modification is the most direct approach to improving bamboo reinforcement bonding performance. Ghavami's early work attempted various treatment methods, including treating bamboo reinforcement with Negrolin (a bitumen-based product), then wrapping with 1.5 mm diameter steel wire and coating with sand, as well as using Sikadur 32 Gel epoxy adhesive coating. Pull-out test results showed that epoxy adhesive treated specimens achieved a 5.29-fold increase in bond strength, with the most significant effect. This finding has been repeatedly verified by subsequent studies, with Kumar et al. and Mengistu both reporting that epoxy-based adhesives can effectively improve the bond strength between bamboo and concrete.

Sand coating is another economically effective surface treatment method. Research by Shah and Rathod (2025) showed that the bond strength between sand-coated bamboo strips and concrete increased by 32% compared to untreated bamboo strips. The mechanism of sand lies in increasing interface roughness, thereby providing stronger mechanical interlocking.

3.2 Geometric Configuration Improvement

Creating semicircular corrugations or tying steel wire on the bamboo reinforcement surface is a geometric method to improve bonding performance. Studies show that corrugation treatment and steel wire binding can increase bond strength by 80% compared to ordinary bamboo reinforcement. The core logic of this strategy is to transform the friction-type bonding between bamboo reinforcement and concrete into mechanical interlocking-type bonding, which is insensitive to changes in bamboo surface conditions and therefore more reliable.

More sophisticated designs have appeared in recent research: drilling holes in bamboo bars and inserting bamboo pins, while coating the bamboo reinforcement with rubber. Test results show that the more bamboo pins, the greater the bond strength, but there exists an optimal density—after approximately 25-50 mm spacing, bond strength no longer significantly increases. The mechanism of rubber coating differs from epoxy coating: rubber inhibits bamboo water absorption and swelling, reducing interface relaxation caused by humidity changes, thereby delaying the occurrence of initial slip. Notably, epoxy resin filling the hollow part of bamboo has minimal effect on maximum bond strength, but can prevent longitudinal crack formation, allowing bond strength to be maintained at larger slip values.

3.3 Effective Range of Bond Length

The influence of bond length on bond strength exhibits nonlinear characteristics. Early research pointed out that the effective anchorage length of bamboo reinforcement should be no less than 305 mm, beyond which bond strength tends to stabilize. Recent experiments confirmed the trend of bond strength decreasing with increasing bond length, which is consistent with the general规律 of steel reinforcement bonding behavior, but bamboo reinforcement shows more significant attenuation. This finding has direct guiding significance for anchorage design of bamboo reinforced concrete members: excessively long anchorage lengths not only fail to improve bonding effectiveness but actually reduce average bond strength due to uneven interface stress distribution.

4. Mechanical Properties of Bamboo Reinforced Concrete

4.1 Compressive Performance

The influence of bamboo reinforcement on concrete compressive performance depends on the form, content, and configuration of bamboo reinforcement. The systematic experiments by Sayeed et al. (2023) revealed several important patterns: when bamboo reinforcement volume content was 0.6% and length-to-diameter ratios were 20 and 30, concrete cube compressive strength increased by 3.24% and 17.33% respectively; while at volume contents of 1.2% and 2.4% with length-to-diameter ratio of 10, compressive strength improvement was approximately 21%. Notably, bamboo reinforcement diameter

also significantly affects compressive performance, but this influence has interactive effects with length-to-diameter ratio.

The influence of bamboo fiber incorporation on concrete compressive strength shows different patterns. Research indicates that low content (1%-3%) bamboo fiber has an improving effect on compressive strength, with Chinchay Vicente (2025) reporting that 3% bamboo fiber content achieved concrete compressive strength of 300.40 kg/cm². However, at high content levels, compressive strength shows a declining trend, which may be related to fiber agglomeration and increased porosity. Carbonation curing can further amplify the reinforcing effect of bamboo fiber, with 1% bamboo fiber combined with carbonation curing achieving cube compressive strength 14.5% higher than water curing conditions.

4.2 Flexural and Tensile Performance

The flexural performance of bamboo reinforced concrete is the core concern for its structural application. Tan's research (cited in Sayeed et al.) reported that the flexural capacity of bamboo reinforced concrete beams is approximately 46% that of steel reinforced concrete beams. This data clarifies the limitation of bamboo reinforcement as main load-bearing reinforcement: at the same reinforcement ratio, the load-bearing capacity of bamboo reinforced beams is less than half that of steel reinforced concrete beams.

However, hybrid reinforcement strategy opens a feasible path for improving the flexural performance of bamboo reinforced concrete. Experiments by Muñoz Bustamante (2025) showed that using a "1 steel bar + 3 bamboo bars" hybrid reinforcement scheme, the beam's flexural efficiency reached 89.00 kgf/cm², achieving over 86% relative efficiency compared to the 103.23 kgf/cm² of purely steel-reinforced beams. This result has important engineering significance: in low-rise buildings, hybrid reinforcement can significantly reduce steel usage while maintaining acceptable structural performance.

The improvement of bamboo fiber on concrete flexural performance is more modest. Systematic reviews indicate that bamboo fiber mainly improves concrete toughness and deformation capacity, with limited improvement in flexural strength. The contribution of fibers is mainly reflected in crack bridging and energy absorption, rather than directly providing tensile load-bearing capacity.

4.3 Deformation and Ductility Characteristics

The load-deformation behavior of bamboo reinforced concrete members has distinctive characteristics. Due to the low elastic modulus of bamboo, members produce significant deflection even at lower load levels, and crack widths are also larger than steel reinforced concrete members. Japanese scholars' research on bamboo reinforced concrete slabs shows that their ultimate flexural strength can be predicted through structural calculations, providing a basis for establishing design methods.

Regarding ductility, seismic performance research on bamboo reinforced concrete columns provides interesting data. Concrete columns repaired using bamboo stirrups (spacing 70 mm) achieved a ductility ratio of 11.7, significantly improved compared to the original column's 7.3. When bamboo stirrup spacing was densified from 110 mm to 70 mm, the ductility ratio increased by approximately 45%-90%. This indicates that bamboo has considerable application potential as transverse confinement reinforcement, with a mechanism similar to steel stirrups, improving member deformation capacity by confining core concrete.

5. Application Prospects and Core Challenges

5.1 Applicable Scenarios

Based on existing research evidence, the reasonable application scope of bamboo reinforced concrete is mainly concentrated in the following areas. First, low-rise buildings and non-seismic critical members, where the lower stiffness of bamboo reinforced concrete can be compensated through section design, while its low-carbon advantages are fully utilized. Second, temporary structures, where the "emergency reinforcement" purpose originally envisioned by Geymayer and Cox still has practical significance today, especially in post-disaster rapid reconstruction and military engineering. Third, hybrid reinforcement members, replacing steel with bamboo reinforcement, achieving a balance between performance and sustainability.

5.2 Durability Challenges

Durability is the most fundamental challenge facing bamboo reinforced concrete. The alkaline environment inside concrete poses a threat to the long-term performance of bamboo, and bamboo's hygroscopic nature causes repeated expansion and contraction under humidity changes, gradually weakening interface bonding. Although rubber coating and epoxy coating can delay this process, long-term performance data remain scarce. Japanese scholars' experimental research on bamboo reinforcement alkali resistance is an important advance in this direction, but the experimental period is insufficient to assess full life-cycle performance.

5.3 Standardization and Design Methods

Bamboo reinforced concrete currently lacks widely accepted design codes and standards. The natural variability of bamboo mechanical properties, performance differences brought by different bamboo species and treatment methods, and lack of long-term performance data all hinder the advancement of standardization work. Geymayer and Cox attempted to propose "temporary" design recommendations in the 1970s, but half a century later, systematic design methods have still not been established. Recent

Japanese research attempts to use structural calculation methods to predict the ultimate strength of bamboo reinforced concrete slabs, which represents an important step toward establishing design methods.

6. Conclusions and Outlook

Research on bamboo reinforced concrete has spanned half a century, accumulating rich knowledge in material treatment, interface modification, and structural performance, but still faces multiple obstacles in moving from laboratory to engineering practice.

Core findings can be summarized as: (1) Bamboo has high tensile strength but low elastic modulus, determining the basic characteristic of bamboo reinforced concrete as "adequate strength, insufficient stiffness"; (2) Weak interface bonding is the core bottleneck, with epoxy coating and sand coating being the most effective improvement methods, increasing bond strength by several times; (3) The flexural capacity of bamboo reinforced concrete beams is approximately 46% that of steel reinforced concrete beams, but hybrid reinforcement can increase relative efficiency to over 86%; (4) Bamboo fiber mainly improves toughness and deformation capacity, with limited improvement in compressive and flexural strength that depends on content and curing conditions; (5) Bamboo stirrups show good ductility improvement effects in column confinement, making it a promising application direction.

Future research should focus on: accumulation of long-term durability data, especially performance evolution of coated bamboo reinforcement under alkaline environments and humidity cycles; systematization of design methods, establishing reliability-based design criteria for bamboo reinforced concrete members; standardization of bamboo species and processing techniques to reduce variability in mechanical properties; and full life-cycle assessment to quantify the carbon emission advantages of bamboo reinforced concrete relative to steel reinforced concrete.

The engineering significance of bamboo reinforced concrete lies not in completely replacing steel reinforcement, but in providing sustainable alternative solutions for specific application scenarios. In the building industry facing increasing carbon emission pressure, the value of this technical route will gradually emerge.

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